

1 A Unified EM Framework for Misclassified Employment Dynamics with Tenure and Nonemployment Durations

1.1 States, sampling, and notation

We observe individuals $i = 1, \dots, N$ at $t \in \{1, 2, 3\}$ (quarters). The *true* employment state is $s_{it}^* \in \{0, 1\}$, where 1 denotes employment. The *observed* state is $s_{it} \in \{0, 1\}$, measured with misclassification. Let the latent 3-period history be

$$H_i = (h_{i1}, h_{i2}, h_{i3}) \in \{0, 1\}^3, \quad h_{it} \equiv s_{it}^*.$$

We write $h = (h_1, h_2, h_3)$ when individual indices are inessential, and denote the set of the eight histories by $\mathcal{H}$.

We also observe *tenure* (employment duration) $g_{it} \geq 0$ and *nonemployment duration* $d_{it} \geq 0$. At any t , only one of (g_{it}, d_{it}) is observed, determined by s_{it} : if $s_{it} = 1$ we observe g_{it} , if $s_{it} = 0$ we observe d_{it} . Survey weights are $w_i > 0$.

1.2 True dynamics: discrete- and continuous-time views

Discrete-time Markov chain. The true chain $\{s_{it}^*\}$ is first-order Markov with

$$\alpha \equiv \Pr(s_{i1}^* = 1), \quad \theta_1 \equiv \Pr(s_{it}^* = 1 \mid s_{i,t-1}^* = 1), \quad \theta_0 \equiv \Pr(s_{it}^* = 1 \mid s_{i,t-1}^* = 0).$$

Hence the prior mass on a history $h \in \mathcal{H}$ is

$$p(H_i = h \mid \alpha, \theta_0, \theta_1) = \alpha^{h_1} (1-\alpha)^{1-h_1} \prod_{t=2}^3 \left[\theta_1^{\mathbf{1}\{h_{t-1}=1, h_t=1\}} (1-\theta_1)^{\mathbf{1}\{h_{t-1}=1, h_t=0\}} \theta_0^{\mathbf{1}\{h_{t-1}=0, h_t=1\}} (1-\theta_0)^{\mathbf{1}\{h_{t-1}=0, h_t=0\}} \right]. \quad (1)$$

Continuous-time embedding and duration parameters. Assume a two-state continuous-time Markov chain with intensities α_1 (job loss, $1 \rightarrow 0$) and α_0 (job entry, $0 \rightarrow 1$). With quarterly spacing $\Delta = 1$ quarter = 3 months, the discrete-time transition probabilities and the continuous-time intensities are linked by

$$\theta_1 = 1 - e^{-\alpha_1 \cdot 3}, \quad \theta_0 = 1 - e^{-\alpha_0 \cdot 3}. \quad (2)$$

The memoryless property implies *true* spell lengths are exponential: $G \sim \text{Exp}(\lambda_g)$, $D \sim \text{Exp}(\lambda_d)$ with $\lambda_g = \alpha_1$ and $\lambda_d = \alpha_0$. Thus, under the *linked* specification,

$$\lambda_g \equiv -\frac{1}{3} \log(1 - \theta_1), \quad \lambda_d \equiv -\frac{1}{3} \log(1 - \theta_0). \quad (3)$$

We allow two alternatives in estimation: (i) *linked* (enforce (3)), or (ii) *semi-linked/free* (estimate λ_g, λ_d directly, optionally regularized toward (3)).

1.3 Deterministic clocks for tenure and nonemployment

Given a history h , the *true* tenure and nonemployment clocks evolve deterministically at quarterly grid points:

$$g_t^*(h) = h_t(g_{t-1}^*(h) + \frac{1}{4}), \quad g_0^* \equiv 0; \quad d_t^*(h) = (1 - h_t)(d_{t-1}^*(h) + \frac{1}{4}), \quad d_0^* \equiv 0. \quad (4)$$

Thus a *continuation* $h_{t-1} = h_t = 1$ increments tenure by 0.25, and a *start* $h_{t-1} = 0, h_t = 1$ resets tenure to 0.25. Analogously for nonemployment.

1.4 Observation model: misclassification and duration measurement

State misclassification. Observed states are misclassified symmetrically with probability π :

$$q_{ab}(\pi) \equiv \Pr(s_{it} = a \mid s_{it}^* = b) = \pi \mathbf{1}\{a \neq b\} + (1 - \pi) \mathbf{1}\{a = b\}, \quad a, b \in \{0, 1\}. \quad (5)$$

Duration measurement: different variances by regime. When $s_{it} = s_{it}^*$ (correct classification), we observe the *appropriate* clock with additive Gaussian noise; at spell *starts* the marginal law is exGaussian (EMG), and at *continuations* we use Gaussian *increments*. We allow different measurement variances in employment and nonemployment:

$$\varepsilon_{it}^g \sim \mathcal{N}(0, \sigma_g^2), \quad \varepsilon_{it}^d \sim \mathcal{N}(0, \sigma_d^2). \quad (6)$$

Matched employment ($s_{it} = h_t = 1$).

- Start ($h_{t-1} = 0, h_t = 1$): g_{it} follows the EMG density,

$$g_{it} = X_{it} + \varepsilon_{it}^g, \quad X_{it} \sim \text{Exp}(\lambda_g),$$

with pdf $f_{\text{EMG}}(g_{it}; \lambda_g, \sigma_g^2)$.

- Continuation ($h_{t-1} = h_t = 1$): the increment

$$\Delta_{it}^g \equiv (g_{it} - g_{i,t-1}) - \frac{1}{4} \sim \mathcal{N}(0, 2\sigma_g^2), \quad (7)$$

independently over t conditional on the history.

Matched nonemployment ($s_{it} = h_t = 0$). Starts are EMG with (λ_d, σ_d^2) , and continuations use

$$\Delta_{it}^d \equiv (d_{it} - d_{i,t-1}) - \frac{1}{4} \sim \mathcal{N}(0, 2\sigma_d^2). \quad (8)$$

Misclassified durations ($s_{it} \neq h_t$). If $h_t = 1, s_{it} = 0$ (true employment, observed nonemployment), the *inappropriate* duration $d_{it} \sim \text{Exp}(\lambda_d)$. If $h_t = 0, s_{it} = 1$, then $g_{it} \sim \text{Exp}(\lambda_g)$. These densities carry identification information about (λ_g, λ_d) even absent starts in the matched block.

1.5 Observed-data likelihood

Let $y_i = (s_{i1}, s_{i2}, s_{i3}, g_{i1}, g_{i2}, g_{i3}, d_{i1}, d_{i2}, d_{i3})$. Conditional on h ,

$$p(y_i | h; \Theta) = \underbrace{\prod_{t=1}^3 q_{s_{it}, h_t}(\pi)}_{\text{state misclassification}} \times \underbrace{\prod_{t=1}^3 \ell_t(y_{it} | s_{it}, h_{t-1}, h_t; \lambda_g, \lambda_d, \sigma_g^2, \sigma_d^2)}_{\text{duration emission}},$$

where ℓ_t selects the EMG/Normal-increment/Exponential factor appropriate to (s_{it}, h_{t-1}, h_t) using (7)–(8). The prior over histories is (1). With survey weights $\{w_i\}$, the observed-data log-likelihood is

$$\ell(\Theta) = \sum_{i=1}^N w_i \log \left[\sum_{h \in \mathcal{H}} p(H_i = h | \alpha, \theta_0, \theta_1) \prod_{t=1}^3 q_{s_{it}, h_t}(\pi) \prod_{t=1}^3 \ell_t(\cdot) \right], \quad (9)$$

with parameters

$$\Theta = \begin{cases} (\alpha, \theta_0, \theta_1, \pi, \sigma_g^2, \sigma_d^2) & \text{(linked: } \lambda\text{s via (3))}, \\ (\alpha, \theta_0, \theta_1, \pi, \sigma_g^2, \sigma_d^2, \lambda_g, \lambda_d) & \text{(semi-linked/free)}. \end{cases}$$

1.6 Complete-data formulation

Introduce indicators $z_{ih} = \mathbf{1}\{H_i = h\}$ with $\sum_h z_{ih} = 1$. The complete-data log-likelihood is

$$\log L_c(\Theta) = \sum_{i=1}^N \sum_{h \in \mathcal{H}} w_i z_{ih} \left\{ \log p(H_i = h | \alpha, \theta_0, \theta_1) + \sum_{t=1}^3 \log q_{s_{it}, h_t}(\pi) + \sum_{t=1}^3 \log \ell_t(\cdot) \right\}. \quad (10)$$

1.7 EM algorithm

Starting from $\Theta^{(0)}$, iterate E- and M-steps until convergence of $\ell(\Theta)$.

E-step. Compute responsibilities (posterior probabilities of histories):

$$\gamma_{ih} \equiv E[z_{ih} | y_i; \Theta^{\text{old}}] = \frac{p(H_i = h | \alpha, \theta_0, \theta_1) \prod_t q_{s_{it}, h_t}(\pi) \prod_t \ell_t(\cdot)}{\sum_{h' \in \mathcal{H}} p(H_i = h' | \alpha, \theta_0, \theta_1) \prod_t q_{s_{it}, h'_t}(\pi) \prod_t \ell'_t(\cdot)}. \quad (11)$$

Accumulate weighted sufficient statistics (with $w_i\gamma_{ih}$ as weights):

$$\begin{aligned}
C_1 &= \sum_{i,h} w_i\gamma_{ih} h_1, & C_0 &= \sum_{i,h} w_i\gamma_{ih} (1 - h_1), \\
D_1 &= \sum_{i,h} w_i\gamma_{ih} \sum_{t=2}^3 \mathbf{1}\{h_{t-1} = 1\}, & D_0 &= \sum_{i,h} w_i\gamma_{ih} \sum_{t=2}^3 \mathbf{1}\{h_{t-1} = 0\}, \\
T_{11} &= \sum_{i,h} w_i\gamma_{ih} \sum_{t=2}^3 \mathbf{1}\{h_{t-1} = 1, h_t = 1\}, & T_{01} &= \sum_{i,h} w_i\gamma_{ih} \sum_{t=2}^3 \mathbf{1}\{h_{t-1} = 0, h_t = 1\}, \\
M &= \sum_{i,h} w_i\gamma_{ih} \sum_{t=1}^3 \mathbf{1}\{s_{it} \neq h_t\}.
\end{aligned} \tag{12}$$

For the duration block, separate matched continuations, matched starts, and misclassified:

$$\begin{aligned}
S_g &= \sum_{i,h} w_i\gamma_{ih} \sum_{t: h_{t-1}=h_t=1=s_{it}} \left(\Delta_{it}^g\right)^2, & N_g &= \sum_{i,h} w_i\gamma_{ih} \sum_{t: h_{t-1}=h_t=1=s_{it}} 1, \\
S_d &= \sum_{i,h} w_i\gamma_{ih} \sum_{t: h_{t-1}=h_t=0=s_{it}} \left(\Delta_{it}^d\right)^2, & N_d &= \sum_{i,h} w_i\gamma_{ih} \sum_{t: h_{t-1}=h_t=0=s_{it}} 1,
\end{aligned} \tag{13}$$

with $\Delta_{it}^g, \Delta_{it}^d$ from (7)–(8). For misclassification exponentials,

$$\begin{aligned}
K_g &= \sum_{i,h} w_i\gamma_{ih} \sum_{t: h_t=0, s_{it}=1} g_{it}, & n_g &= \sum_{i,h} w_i\gamma_{ih} \sum_{t: h_t=0, s_{it}=1} 1, \\
K_d &= \sum_{i,h} w_i\gamma_{ih} \sum_{t: h_t=1, s_{it}=0} d_{it}, & n_d &= \sum_{i,h} w_i\gamma_{ih} \sum_{t: h_t=1, s_{it}=0} 1.
\end{aligned} \tag{14}$$

Finally, collect matched *starts* (for EMG), e.g.

$$\mathcal{S}_g = \{(i, t, h) : h_{t-1} = 0, h_t = 1, s_{it} = 1\}, \quad \mathcal{S}_d = \{(i, t, h) : h_{t-1} = 1, h_t = 0, s_{it} = 0\}, \tag{15}$$

and the corresponding weighted samples g_{it} or d_{it} with weights $w_i\gamma_{ih}$.

M-step. Maximize $Q(\Theta \mid \Theta^{\text{old}}) = E[\log L_c(\Theta) \mid \{y_i\}; \Theta^{\text{old}}]$, which amounts to replacing z_{ih} by γ_{ih} in (10), yielding the following updates.

Markov block (closed form):

$$\hat{\alpha} = \frac{C_1}{C_0 + C_1}, \quad \hat{\theta}_1 = \frac{T_{11}}{D_1}, \quad \hat{\theta}_0 = \frac{T_{01}}{D_0}. \tag{16}$$

If stationarity is imposed, set $\alpha \leftarrow \theta_0/(\theta_0 + 1 - \theta_1)$.

Misclassification (closed form):

$$\hat{\pi} = \frac{M}{3 \sum_i w_i}. \tag{17}$$

(Optionally shrink π via a Beta(a_π, b_π) prior: replace M by $M + a_\pi - 1$ and the denominator by $3 \sum_i w_i + a_\pi + b_\pi - 2$.)

Measurement variances in continuations (closed form):

$$\hat{\sigma}_g^2 = \frac{S_g}{2N_g}, \quad \hat{\sigma}_d^2 = \frac{S_d}{2N_d}. \quad (18)$$

(The factor 2 reflects $\text{Var}(\Delta) = 2\sigma^2$. Enforce small floors if needed.)

Exponential rates. Two alternatives:

Linked (CTMC) model: set λ_g, λ_d by (3) using $(\hat{\theta}_1, \hat{\theta}_0)$. No separate M-step is needed.

Semi-linked/free model: use misclassification draws and matched starts.

- From misclassification: $\hat{\lambda}_g^{(\text{mis})} = n_g/K_g$, $\hat{\lambda}_d^{(\text{mis})} = n_d/K_d$.
- From matched starts (EMG): either (A) *maximize* the EMG contribution numerically in (λ_g, σ_g^2) and in (λ_d, σ_d^2) using weighted samples $\mathcal{S}_g, \mathcal{S}_d$; or (B) *augment* by latent exponentials at starts (call them X_{it} and Y_{it}) and set

$$\hat{\lambda}_g^{(\text{start})} = \frac{\sum_{(i,t,h) \in \mathcal{S}_g} w_i \gamma_{ih}}{\sum_{(i,t,h) \in \mathcal{S}_g} w_i \gamma_{ih} E[X_{it} \mid g_{it}, \lambda_g, \sigma_g^2]}, \quad \hat{\lambda}_d^{(\text{start})} = \frac{\sum_{(i,t,h) \in \mathcal{S}_d} w_i \gamma_{ih}}{\sum_{(i,t,h) \in \mathcal{S}_d} w_i \gamma_{ih} E[Y_{it} \mid d_{it}, \lambda_d, \sigma_d^2]},$$

where the EMG conditional expectations have closed form via Mills ratios.

Combine the two sources (e.g. inverse-variance weighting or joint maximization of their summed log-likelihoods). If one prefers to softly *link* λ s to θ s, add a quadratic penalty $\rho(\lambda_g + \frac{1}{3} \log(1 - \theta_1))^2 + \rho(\lambda_d + \frac{1}{3} \log(1 - \theta_0))^2$.

Iteration and convergence. Iterate (E-step, M-step) until $\ell(\Theta)$ in (9) increases by less than a tolerance. The E-step uses (11); the M-step applies (16)–(18) and the chosen update for (λ_g, λ_d) . Monotone ascent is guaranteed if each M-step maximizes Q .

1.8 Identification and practical considerations

- *Joint identification of durations and transitions.* The misclassification exponentials identify (λ_g, λ_d) even absent starts; matched *starts* (EMG) reinforce identification. Linking λ s to θ s via (3) ties the duration and transition views of the same process.
- *Different measurement variances.* Allowing $\sigma_g^2 \neq \sigma_d^2$ captures regime-specific noise and reduces the temptation to attribute regime asymmetries to π or θ s.
- *Increment likelihoods.* Using (7)–(8) avoids double-counting the same Gaussian noise across consecutive periods and stabilizes estimation.

- *Regularization.* Small floors on σ_g^2, σ_d^2 and weak Beta priors for π guard against pathological solutions typical in mixture-like settings. Transition caps (e.g., $\theta_1 \leq 0.995$) can be imposed if needed.
- *Weights.* All sufficient statistics are weight-adjusted via $w_i \gamma_{ih}$; likewise, the observed-data log-likelihood (9) uses weights.

1.9 Summary of parameterizations

Block	Update/Link
$(\alpha, \theta_0, \theta_1)$	Closed form (16) (stationary α optional).
π	Closed form (17) (optional Beta prior).
(σ_g^2, σ_d^2)	Continuation increments (18) (with floors).
(λ_g, λ_d)	Linked: (3); Free: misclass+EMG (A/B).